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Smart WATER SERVING ROBOT

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With the advent of artificial intelligence, robots find application in various sectors such as industry, homes, bars, restaurants, events, agriculture, and even space. In light of the broad scope of robot applications, this smart water serving robot was developed. While there are several water-serving robots on the market, many are costly and complex in design and heavy in weight.

This economical and lightweight water-serving robot addresses this issue with a simple design. It demonstrates high precision in its control.

The author's prototype is shown in Fig. 1. The components required for this device are listed in Bill of Materials table.

Circuit and working

The circuit diagram of the smart water serving robot is shown in Fig. 2. It is built around Arduino Uno (MOD1), Bluetooth HC-05 (MOD 2), L298 motor driver, and a few other components.

BILL OF MATERIALS

Components	Quantity
Arduino Uno (MOD1)	1
Bluetooth HC-05 (MOD2)	1
L298 motor driver	1
12V, 500-RPM geared DC motor	2
Battery module 18650, 3.7V 2000mAh	3
Battery holder	1
Breadboard	1
Jumper wires	20
Hardware parts such as 100cm diameter 4 tyres, wooden planks, robot chassis and screws	

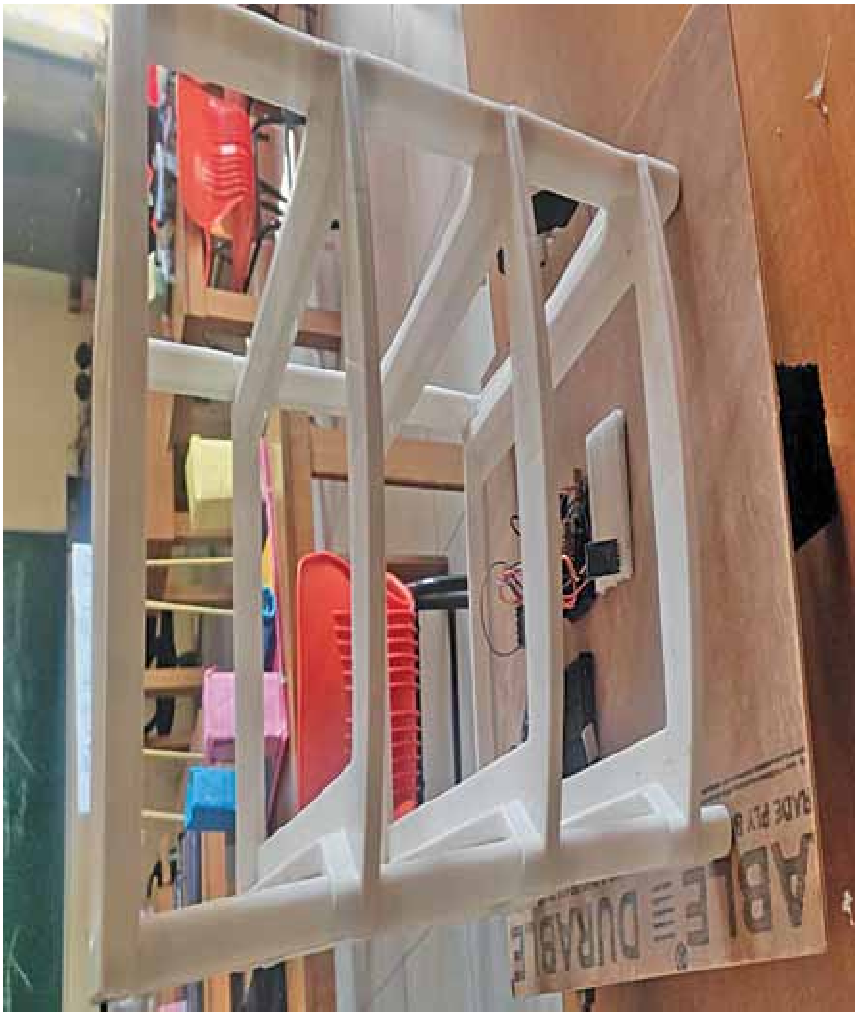

Fig. 1: Author's robot for serving

Connect the V_{CC} in the circuit diagram to the positive terminal of the 12V battery and GND to the negative terminal. For this setup, the author used a battery holder and three 3.7V batteries. This totals to 11 to 12V input for the motors (3.7Vx3 = 11.1V). The Arduino utilises the Vcc pin to convert this to 5V using its built-in voltage regulator. Alternatively, a single 12V battery can be used.

The pins of the HC-05 Bluetooth module, such as Vcc, RX, TX, and GND, are connected to the corresponding Arduino Uno pins of Vcc, TX, RX, and GND, respectively. Additionally, Arduino Uno pins D9, D10, D7, D8, D11, and D12 are connected to enable pins A, B, IN1, IN2, IN3, and IN4 of the L298 motor